

Polina Kliuchnikova

ML Researcher | RLHF, Game Theory & Applied Mathematics

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Research: istina.msu.ru/workers/251795028/ | Earlier publications under Polina S. Tanasevich

RESEARCH PROFILE

PhD student in AI & Machine Learning at ITMO University and Staff ML Engineer at Huawei. Research spans game-theoretic models of human-AI interaction, cognitive modelling and reinforcement learning. Combine mathematical modelling with hands-on LLM post-training and evaluation.

SELECTED RESEARCH

Game-theoretic models of human oversight

- Modelled evaluator-LLM incentives as a 5×5 strategic-form game; computed mixed-strategy Nash equilibria by support enumeration and analysed best responses and replicator dynamics.
- Studied delegation versus verification in a 2×2 human-AI role game using Nash and logit quantal response equilibria (QRE). Analysed sensitivity across 2,772 model variants and investigated structural non-identifiability of payoff weights using algebra and parameter-recovery simulations.

Cognitively informed reinforcement learning

- Coauthored work integrating a cognitive critic into reinforcement learning from AI feedback (RLAIF) for code generation; compared supervised fine-tuning, contrastive learning and reward models, including Markov-chain reward aggregation. Article in press.

PROFESSIONAL EXPERIENCE

Huawei | Staff ML Engineer | Sep 2025-Present

- Developed RLHF post-training pipelines with PPO, reward modelling, synthetic data generation and training-data quality controls.
- Fine-tuned and post-trained LLMs using multi-GPU DeepSpeed and mixed precision; deployed inference with vLLM. Built evaluation workflows and benchmarks for generated outputs.

Sberbank | Tech Lead, Machine Learning | Feb 2024-Aug 2025

- Led ML engineering work across LLM-based geo-analytics and automated log analysis, taking models from research to deployment; optimised GPU inference using TensorRT and reduced precision.

SELECTED PUBLICATIONS

Published and in-press work are distinguished below. Titles of Russian-language abstracts are translated into English.

In press

S. V. Kovalchuk, D. V. Fedrushkov, P. S. Kliuchnikova and A. T. S. Ireddy. Adaptation of Artificial Intelligence Agents to Individual Operators's Cognitive States Using Reinforcement Learning. In press.

Published

M. Ya. Yakovlev, P. S. Tanasevich, A. V. Vershinin and V. A. Levin. Numerical analysis of the effective thermal properties and the stability for NTE metamaterials using CAE fidesys. AIP Conference Proceedings 2509, 020210 (2022).
doi.org/10.1063/5.0084835

P. S. Tanasevich, M. Ya. Yakovlev and A. V. Vershinin. Solving the inverse problem of estimating effective properties of auxetic metamaterials using machine learning algorithms. XIII All-Russian Congress on Theoretical and Applied Mechanics, pp. 822-824 (2023). Conference abstract; in Russian.

EARLIER EXPERIENCE & MATHEMATICAL RESEARCH

VK / MY.GAMES | Senior Data Analyst | Oct 2021-Sep 2023

Developed lifetime-value and retention models; automated analytical data pipelines with Spark and Airflow and supported product experimentation.

CAE Fidesys | C++ Core Developer | Sep 2020-Aug 2021

Implemented computational mechanics algorithms for effective material properties. Coauthor of two registered software programs: RU 2020665248 and RU 2020667788 (2020).

Lomonosov Moscow State University | Researcher | Dec 2019-Aug 2021

Research in computational mechanics and numerical modelling of heterogeneous materials: finite-element analysis, thermoelasticity, effective properties and stability of metamaterials.

EDUCATION & ADDITIONAL TRAINING

Doctoral studies, AI & Machine Learning (in progress) | ITMO University

MSc, Applied Mathematics | Lomonosov Moscow State University | 2021-2023

BSc, Applied Mathematics | Lomonosov Moscow State University | 2017-2021

Large Language Model Agents MOOC | UC Berkeley RDI | Sep 2024-Jan 2025. Certificate of completion.

METHODS & TOOLS

Game theory: strategic-form games, Nash equilibria, support enumeration, quantal response equilibrium, best-response analysis, replicator dynamics, comparative statics.

Mathematics & statistics: numerical methods, finite elements, inverse problems, Bayesian updating, Markov chains, Monte Carlo simulation, permutation tests, profile likelihood, identifiability analysis.

ML & engineering: Python, C++, PyTorch, RLHF/RLAIF, PPO, SFT, reward modelling, evaluation design, DeepSpeed, vLLM, TensorRT, Docker, Linux, SQL, Spark, Airflow.